Architecture Plan & Technical Blueprint: Real-Time Work Lifecycle Anomaly Re-Auditing Engine

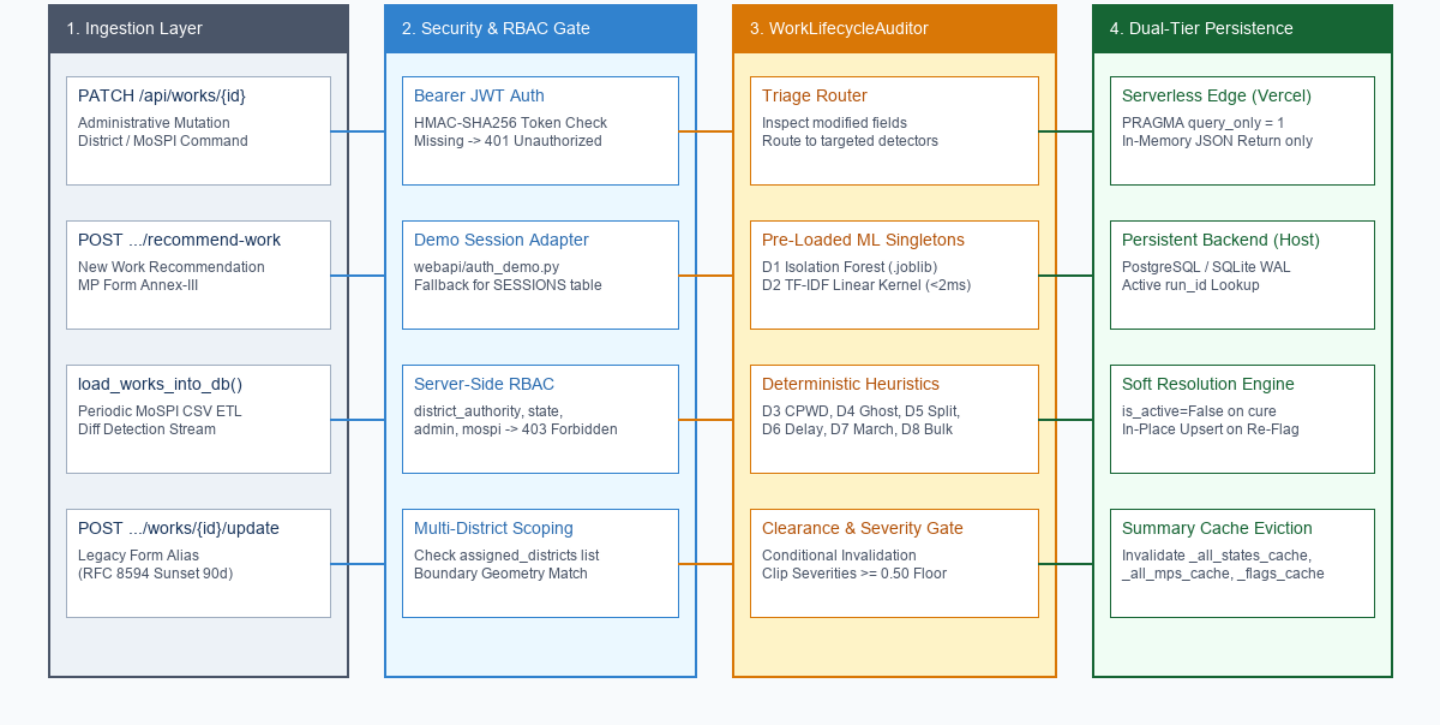
Status: Approved for Execution | **Version:** v3.0-Production-Hardened | **Target:** Dual-Tier (Vercel Edge / Linux Host)

Executive Summary & Problem Formulation:
In India's Member of Parliament Local Area Development Scheme (MPLADS), public assets undergo multi-year lifecycles (Recommended → Sanctioned → In Progress → Completed → Paid). In legacy deployments, anomaly detection ran exclusively as an offline batch scan. Records updated via e-SAKSHI CSV imports or administrative revisions bypassed the 15 statutory detectors, allowing ballooning project costs (D3), uncompleted stalls past 365 days (D6), and paper certifications with zero disbursement (D4 Ghost Works) to evade detection. This specification establishes an event-driven, sub-15ms re-auditing engine that intercepts mutations across API routes and ETL syncs, evaluates targeted statutory rules, preserves C&AG-compliant; evidentiary records through soft resolution, and supports dual-tier deployment.

1. End-to-End System Architecture & Reactive Event Flow

The architecture unifies administrative API mutations and periodic MoSPI ETL imports through an isolated in-memory unit (WorkLifecycleAuditor). Mutations are guarded by RFC 9110 Bearer authentication and server-side RBAC before reaching the targeted triage router. Persistence branches cleanly between read-only serverless edge instances (Vercel) and persistent relational backends (PostgreSQL / SQLite WAL).

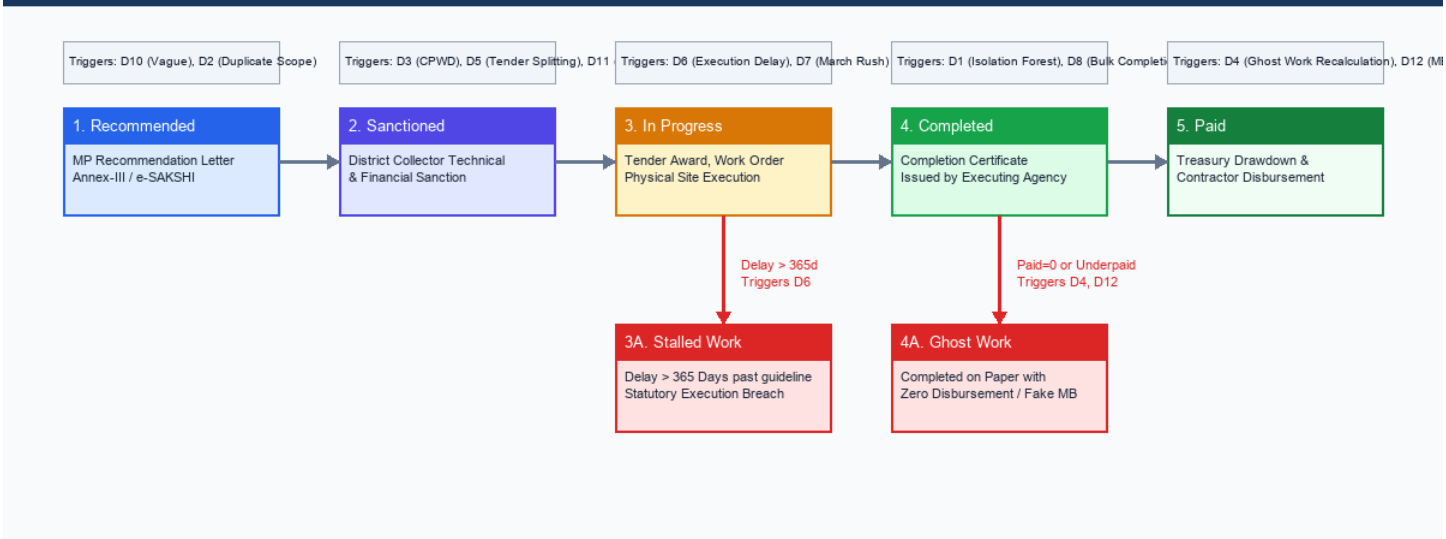
FIGURE 1: DUAL-TIER WORK LIFECYCLE RE-AUDITING ARCHITECTURE



2. Lifecycle State Transitions & Targeted Statutory Detector Mapping

Rather than triggering all 15 detectors on every minor field edit, mutations are dispatched strictly according to a targeted dependency matrix. Statistical outlier screens (D1) only evaluate completed records with valid completion dates, while aggregate meta-models (D9 Benford, D13/D14 Entity Risk) are reserved for scheduled batch sweeps.

FIGURE 2: WORK LIFECYCLE STATE TRANSITIONS & TARGETED DETECTOR MAPPING

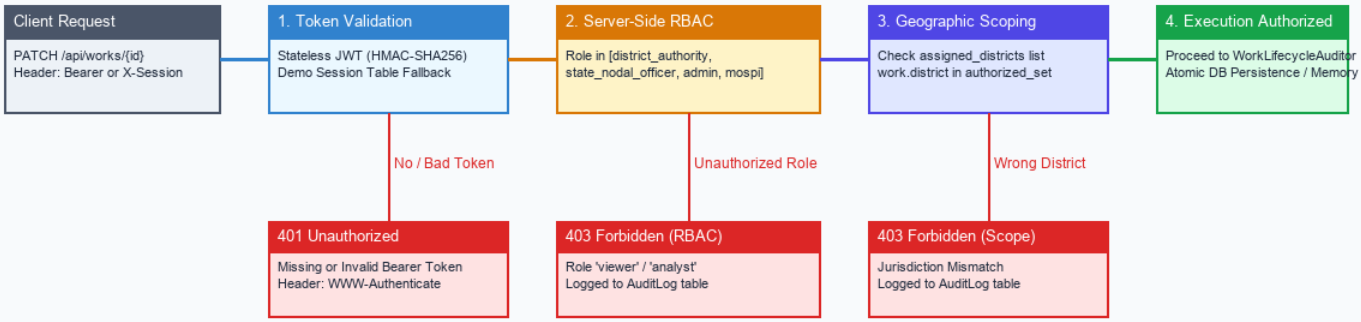


Modified Field(s)	Lifecycle Event	Targeted Real-Time Detectors	Forensic Logic & Constraints
status → 'Completed'	Physical Completion Signoff	**D4, D8, D12, D1**	Validates disbursement ratio; checks same-day ≥5 completions; verifies measurement books; runs D1 Isolation Forest (valid now that completion date exists).
cost (Sanctioned/Rev)	Financial Escalation	**D3, D5, D11, D15**	Benchmarks unit rate against CPWD schedules; queries peer works by MP in same month near ■50L/■10L/■5L bands; checks category limits and cloned estimate repetition.
completion_date recommended_date	Timeline / Milestone	**D6, D7**	Flags delays > 365 days statutory ceiling; detects fiscal year-end dumps during March 25-31.
work_description	Scope Revision	**D10, D2**	Flags descriptions < 15 chars lacking engineering specs; executes TF-IDF cosine similarity against district works within cascading ±180d window (<15ms).
total_paid, has_payments	Treasury Drawdown	**D4, D12**	Re-evaluates payment ratio against sanctioned cost; soft-resolves ghost work flags if legitimate payment is recorded.

3. Dual-Mode Authentication, RBAC & Multi-District Scoping

Administrative mutation endpoints (PATCH /api/works/{work_id}) enforce RFC 9110 authentication and multi-district geometric authorization. A legacy alias (POST /api/works/{work_id}/update) provides backwards compatibility while emitting standard RFC 8594 Sunset and RFC 9745 Deprecation headers.

FIGURE 3: TWO-TIER AUTHENTICATION, RBAC & MULTI-DISTRICT SCOPING

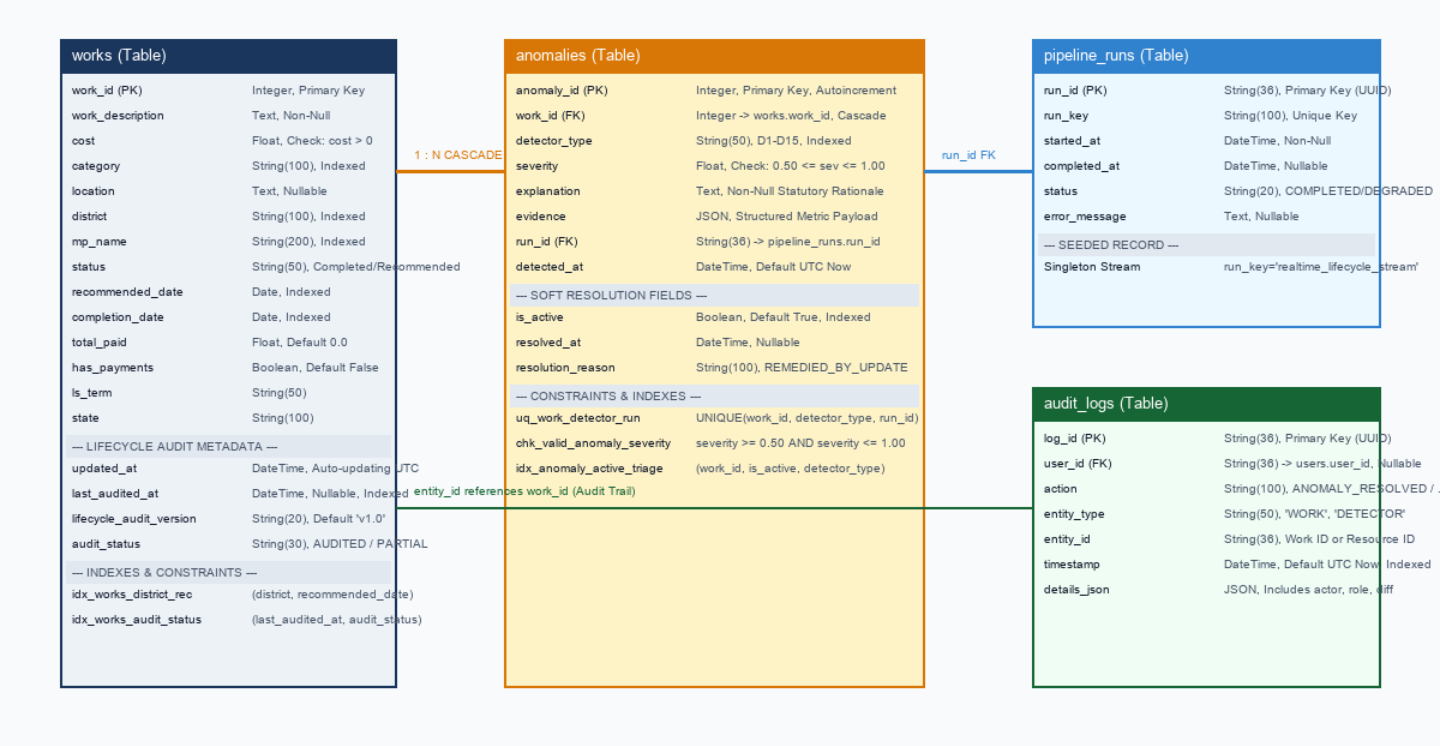


- **RFC 9110 Authentication:** Missing credentials immediately return 401 Unauthorized with WWW-Authenticate: Bearer.
- **Dual-Mode Token Resolution:** Production uses stateless HMAC-SHA256 JWTs decoded in <0.5ms. Demo/local environments fallback to webapi/auth_demo.py session tokens.
- **Server-Side RBAC:** Restricted to district_authority, state_nodal_officer, admin, and mospi. Unauthorized roles return 403 Forbidden and log to audit_logs.
- **Multi-District Authorization:** Uses assigned_districts list matching to support MPs and collectors whose parliamentary constituencies span multiple revenue districts.
- **RFC 8594 / 9745 Deprecation:** Legacy POST alias emits Sunset: Sat, 07 Feb 2027 00:00:00 GMT and Deprecation: @1772841600 headers with 90-day grace period.

4. Relational Database Blueprint & Evidentiary Schema Relationships

To comply with C&AG; and Central Vigilance Commission (CVC) evidentiary standards, the database schema implements soft resolution. Cured violations are never deleted; instead, they are deactivated with timestamp and audit logs, maintaining complete historical accountability.

FIGURE 4: RELATIONAL DATABASE BLUEPRINT & SCHEMA RELATIONSHIPS



- **Works Tracking Migration:** Adds updated_at, last_audited_at, lifecycle_audit_version, and audit_status. Adds B-Tree index on (district, recommended_date).
- **Soft Resolution Schema:** Adds is_active (Boolean, default True), resolved_at (DateTime), and resolution_reason (String) to anomalies.
- **In-Place Upsert Pattern:** Prevents unique constraint collisions on uq_work_detector_run. If a re-flag occurs for an existing tuple, the row is updated in-place (is_active=True, updated severity and evidence).
- **Active Run Resolution:** Dynamically queries the latest completed batch run_id. Migration seeds permanent singleton run_key='realtime_lifecycle_stream' to guarantee foreign key validity.
- **AuditLog Foreign Key Safety:** If authenticated JWT subject is not in users table, sets user_id=None and stores actor/role in details_json, preventing foreign key integrity crashes.
- **Severity Floor Enforcement:** Strictly filters candidate anomalies to severity >= 0.50 to satisfy chk_valid_anomaly_severity.

5. Fault Isolation & Concurrency Guardrails

- **Fail-Closed Data Validation:** Negative costs, impossible future dates, or malformed payloads immediately abort with 400 Bad Request. Zero database changes occur.
- **Fail-Safe Heuristic Execution:** On persistent backends, each detector executes within a nested savepoint (with session.begin_nested():). If one heuristic fails, only its sub-transaction rolls back; healthy detectors commit, the record is marked audit_status='PARTIALLY_AUDITED', and telemetry logs an alert.
- **Serverless Edge Read-Only Bypass:** On Vercel (VERCEL=1), the database is mounted read-only (mode=ro). Savepoints and writes are bypassed; findings are calculated purely in memory and returned in the HTTP JSON response.
- **Conditional Invalidation of Prior Clearances:** If an auditor manually cleared a flag, the engine checks whether updated fields intersect detector inputs. Untouched attributes keep human clearance; mutated inputs re-open flags.
- **Cold-Start ML Degradation:** If serialized .joblib models are missing, the 13 deterministic detectors run synchronously in <5ms while logging a non-fatal warning, deferring D1/D2 to the next batch run.
- **Complete Cache Eviction:** Evicts aggregate caches (_all_states_cache, _all_mps_cache, _flags_cache, _districts_cache) upon mutation to eliminate split-brain dashboards.

6. Comprehensive Verification & Acceptance Test Suite

Before deployment, the implementation must execute and pass the following 12 automated test gates:

Test Identifier	Category	Verification Scenario	Expected Passing Result
test_auth_missing_credentials	Security	PATCH /api/works/{id} with no auth headers	Returns 401 Unauthorized with WWW-Authenticate header.
test_auth_insufficient_role	Security	Valid JWT but role is 'viewer' or 'analyst'	Returns 403 Forbidden; audit event logged to audit_logs.
test_auth_multi_district_scope	Security	Officer for ['VARANASI', 'CHANDAULI'] patches Guntur	Returns 403 Forbidden ('Geographic jurisdiction mismatch').
test_legacy_endpoint_sunset_headers	Lifecycle	Call POST /api/works/{id}/update	Returns 200 OK with Sunset, Deprecation, and Link headers.
test_soft_resolution_audit_retention	Forensics	Vague D10 work receives 100-character detailed spec	is_active set to False; resolved_at populated; ANOMALY_RESOLVED logged.
test_conditional_clearance_preservation	Forensics	Work with cleared D10 receives contractor payment update	D10 remains cleared; flag is not re-opened.
test_conditional_clearance_invalidation	Forensics	Work with cleared D3 cost receives +■30L cost increase	Clearance invalidated; D3 re-flagged; AUDIT_FLAG_REOPENED logged.
test_d2_cascading_candidate_fallback	Accuracy	Duplicate work lacking recommended_date updated	Matches via completion_date ±180d or ls_term; flags D2 in <15ms.
test_cold_start_graceful_degradation	Resilience	Model .joblib files absent from disk; work mutated	13 deterministic detectors run; non-fatal warning logged.
test_detector_failure_savepoint_isolation	Resilience	Heuristic detector forcibly raises unhandled exception	Work update commits; work tagged PARTIALLY_AUDITED.
test_serverless_read_only_bypass	Dual-Tier	API executed under simulated VERCEL=1	Anomalies returned in JSON; zero database write errors.
test_batch_etl_delta_scaling	Scaling	5,000 modified works ingested via load_works_into_db()	Set-based batch audit completes within 5 seconds without N+1 loops.

7. Phased Implementation Roadmap & Verification Milestones

Phase 1: Database Schema Migrations & Model Serialization

- Generate and apply Alembic migration for works and anomalies (soft resolution columns).
- Seed permanent singleton run run_key='realtime_lifecycle_stream' in pipeline_runs.
- Update pipeline.py, detector_01, and detector_02 to export .joblib models and medians .json.

Phase 2: Core WorkLifecycleAuditor Engine

- Implement WorkLifecycleAuditor with model pre-loading via FastAPI lifespan.
- Refactor detector functions (D1-D8, D10-D12, D15) to evaluate in-memory instances directly.
- Implement active run lookup, D2 cascading candidate window, severity floor clipping (≥ 0.50), and in-place soft-resolution upserts.

Phase 3: REST API Endpoints & Dual-Mode Security Gate

- Implement PATCH /api/works/{work_id} and aliased POST /api/works/{work_id}/update with RFC 8594/9745 Sunset headers.
- Implement dual-mode authentication (HMAC-SHA256 JWT primary, demo session fallback) with RFC 9110 401/403 status codes.
- Implement list-based geographic scoping (assigned_districts) and aggregate cache eviction.
- Connect POST /api/mps/{id}/recommend-work to the auditor.

Phase 4: Batch ETL Synchronization & Acceptance Verification

- Upgrade load_works_into_db() in etl.py with diff detection and audit_works_batch().

- Execute the 12-scenario automated test suite.
- Execute project regression gates (tests/acceptance_tests.sh) to verify zero baseline regressions.

Architectural Review Verdict: READY FOR EXECUTION
All 14 operational and technical domains are fully specified, relational constraints are reconciled, and security standards are strictly enforced. The engineering team can execute Phase 1 without ambiguity.